Learning Simple Test-Time Environments for LLM Web Agents

Junxuan Li, Zijun Liu, Ziyi Huang, Peng Li, Yuzhou Liu, Ming Yan, Yang Liu

Paper version: arXiv 2608.29305v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *An agent that succeeds in individual sub-environments may still collapse in a cluttered interface.*

TTED splits each observation into a **sub-goal** and a **sub-observation** and learns without labels.

Motivation: three open gaps

1. **Clutter.** CompWoB+, Qwen3-8B: 85% falls to 19% (3 to 8 components).
2. **Weak exploration.** Cluttered pages lower exploration success.
3. **Noisy judging.** Self-assessment degrades too.

Evaluation setup

CompWoB+ · 600 tasks | WebArena subset · 561 | WorkArena (transfer) | Qwen3-8B base | GPT-4o-mini (ICL) | 3200 sub-envs

- **All online** task success; no offline step accuracy.
- **Held out:** WorkArena only. WebArena is adapted on and tested on the same tasks.

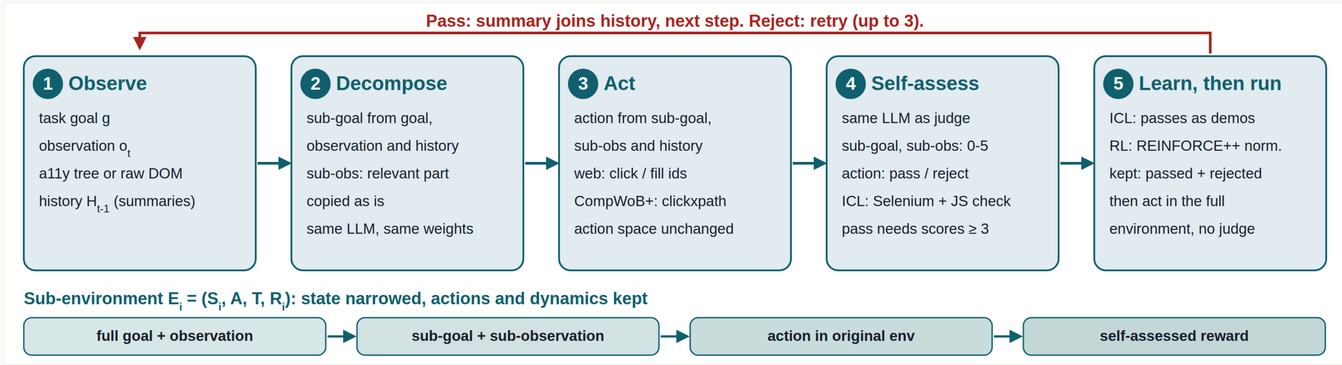
Our read (CUA Radar)

69	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; WorkArena transfer; matched budgets; cost; ablation; human check of reward labels.
- **Flags:** judge not checked for false positives / negatives; adapts on the tasks it is tested on.
- Pattern C13. Variance, code: unclear in record. Authors' numbers.

Method: TTED, two stages around the full task

- **Sees / acts:** a11y tree (WebArena) or raw DOM (CompWoB+) in; BrowserGym actions or clickxpath / type out.
- **New:** each step gets a sub-goal and a pruned sub-observation; the policy acts on those.
- **Signal:** the same LLM scores sub-goal, sub-observation and action; ICL keeps passes, RL trains on all.



Our redraw (not the paper's figure): trial steps with self-assessment, then ICL or RL, then the full environment.

$$\pi^{*}_{\mathrm{TTED}} = \arg\max_{\pi} \sum_{i=1}^{n} \mathbb{E}_{\tau^{(i)}\sim\pi}\left[R^{(i)}(\tau^{(i)})\right]$$

Optimize rewards summed over n local sub-environments

Key results: online success rate (%)

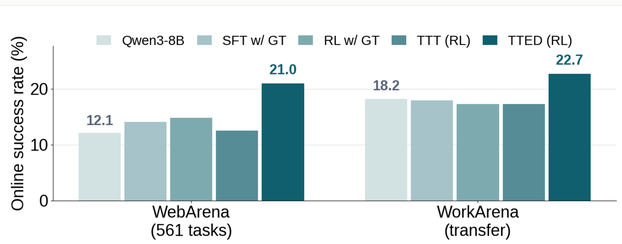


Table 1, redrawn: Qwen3-8B, no further training on WorkArena.

WebArena 12.1 → 21.0; WorkArena 18.2 → 22.7

SO WHAT → *TTRL: 13.5 on WebArena, no WorkArena entry.*

Ablation: WebArena success rate (%)

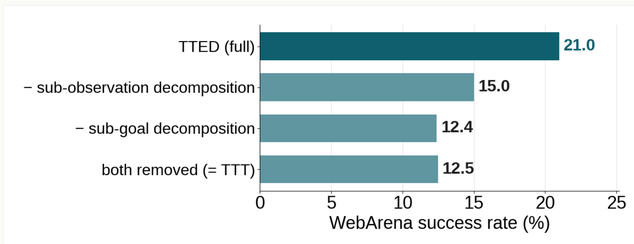


Table 4, redrawn: same RL budget, one decomposition removed at a time.

SO WHAT → *Removing sub-goal decomposition costs most (21.0 to 12.4). Removing sub-observation alone gives 15.0; removing both gives 12.5, the same as TTT. All rows use the same RL budget.*

Headline Numbers

21.0%
WebArena success
vs 12.1% base, no adaptation

- **22.7%** WorkArena (transfer)
- **14.8%** RL w/ ground truth
- **19.4%** with 800 sub-envs

The paper's own controls

CONTROL	TTED	COMPARED WITH
WebArena, 800 sub-envs	19.4	RL w/ GT 14.8 (3200)
Human agreement, 82 tasks	0.85	TTRL 0.68 / TTT 0.71
Qwen3-14B, WebArena	22.5	RL w/ GT 20.0 / TTT 18.7
CompWoB+, 8 comps, ICL	61 / 88	no TTED 19 / 46
Prompt tokens / steps	3948.2 / 8.29	base 6338.9 / 4.72

Takeaway

Splitting the observation lifts Qwen3-8B from 12.1% to 21.0% on WebArena without labels, and to 22.7% on WorkArena; the judge is the agent itself.

14.80 vs 14.40 GPU h (TTT) | 8.29 vs 4.72 steps

Scan to Read

Paper | Code